

"RPI_Robot_HAT" is a raspberry PI HAT designed for small/medium size robots. It has an IMU, Dynamixel connectors and audio. Its dimensions fits a raspberry PI zero

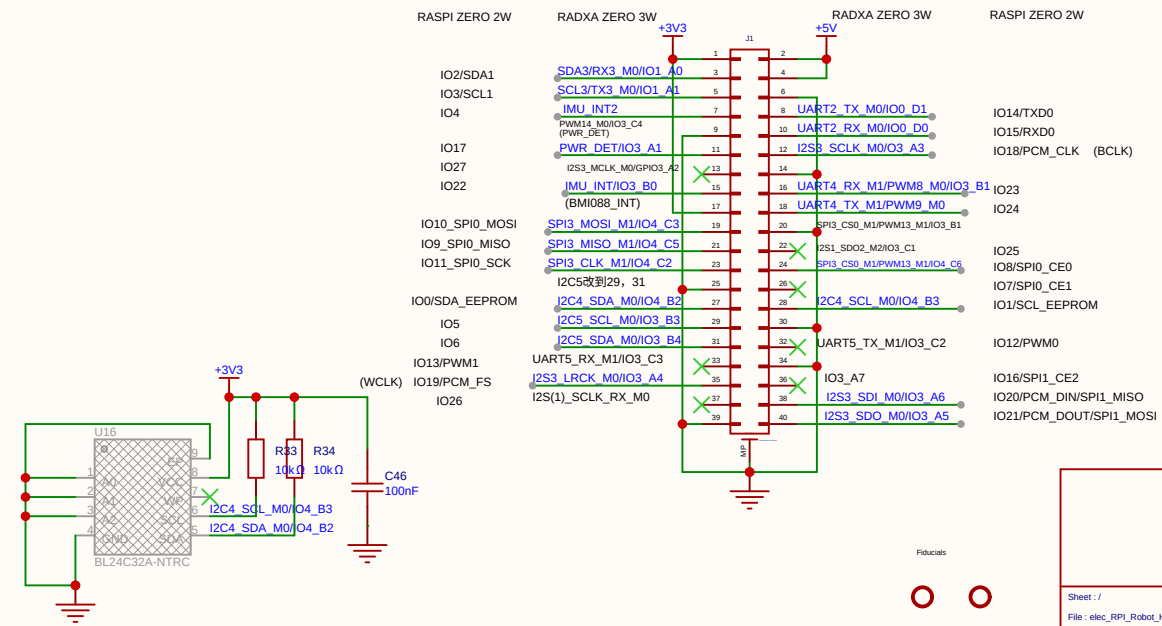
Release	Date	Designer	Check	Comments
A1	06/03/2026	EB	SN	Initial drawing
B1	06/05/2026	EB	SN	[fixed] Acc. is at 0x19 I2C address [fixed] Soft reset of the audio DAC [fixed] Ideal diode [fixed] Know position of IMU [update] Dynamixel_dir is based on Tx [added] IMU can be disconnected [added] IMU_IT is cabled [added] Access to I2C0 [added] Switch off detection
C1	08/07/2026	EB	SN	[fix] 3V3 on IO10/11 [fix] MOS direction of Ideal diode [update] I2C3 is by default instead of reserved I2C

Cabled buses on Raspberry Pis

- Pi zero 2W:
 - I2C0 (pin 27/28 ID_SDC); reserved, but can be on 36 with 0R
 - I2C1 (IO 2/ 3 - pin 3/ 5 SDA/SCL); IMU + Audio + Qwlic 35
- Pi 4:
 - same I2C0/1
 - I2C3 (IO 4/ 5 - pin 7/29 SDA/SCL); Qwlic 36
 - I2C4 (IO 8/ 9 - pin 24/21 SDA/SCL); Qwlic 37
 - I2C5 (IO 19/11 - pin 19/23 SDA/SCL); Qwlic 38
- Pi 5:
 - same I2C0/1
 - I2C2 (IO 2/ 3 - pin 3/ 5 SDA/SCL); Qwlic 35

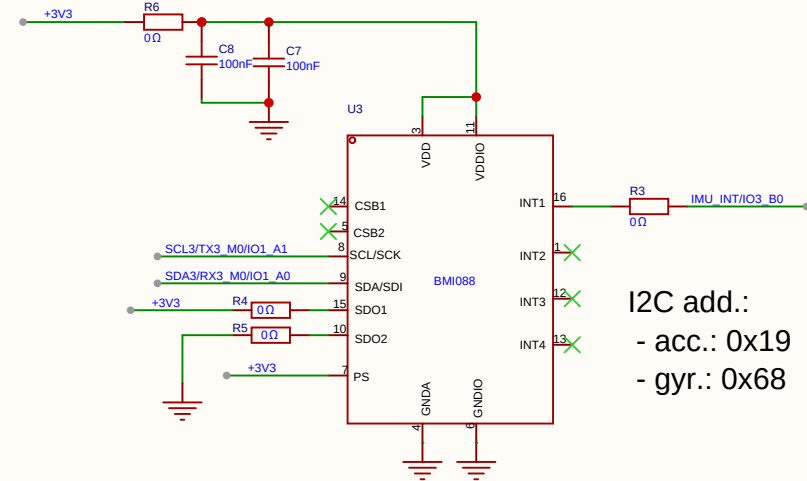
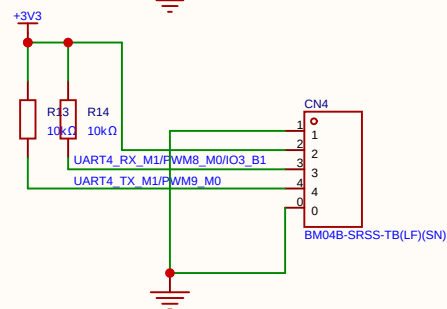
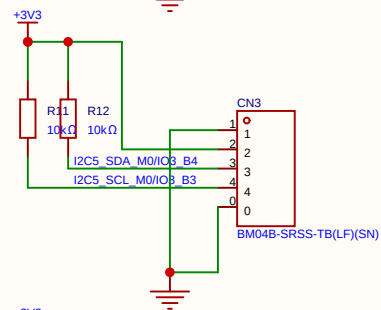
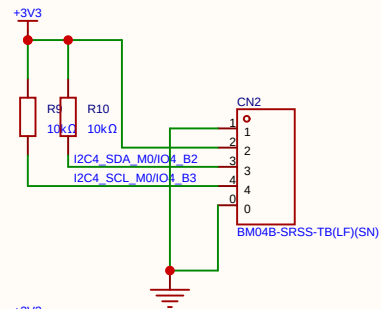
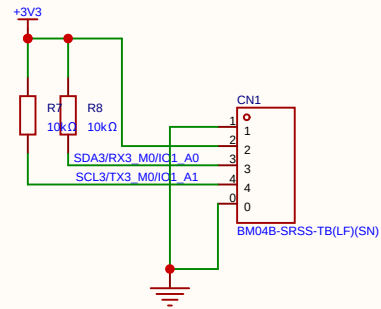
Cabled GPIOs on the 40-way Header:

- GPIO00/01: HAT identification over I2C (reserved)
- GPIO02: IMU + Audio + Stemma/QT - SDA
- GPIO03: IMU + Audio + Stemma/QT - SCL
- GPIO04: Stemma/QT - SDA
- GPIO05: Stemma/QT - SCL
- GPIO06: Switch off detection
- GPIO07
- GPIO08: Stemma/QT - SDA
- GPIO09: Stemma/QT - SCL
- GPIO10: Stemma/QT - SDA
- GPIO11: Stemma/QT - SCL
- GPIO12
- GPIO13:
- GPIO14: Dynamixel - UART_Tx
- GPIO15: Dynamixel - UART_Rx
- GPIO16
- GPIO17
- GPIO18: Audio, PCM Clk
- GPIO19: Audio, PCM FS
- GPIO20: Audio, Din
- GPIO21: Audio, Dout
- GPIO22: IMU_IT
- GPIO23
- GPIO24
- GPIO25
- GPIO26
- GPIO27

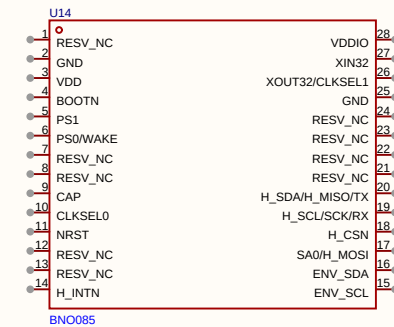


Sheet : /	
File : elec_RPI_Robot_HAT	
Title :	
Size : "A4"	Date :
Rev :	
Id : /	

I2C SENSOR BUS

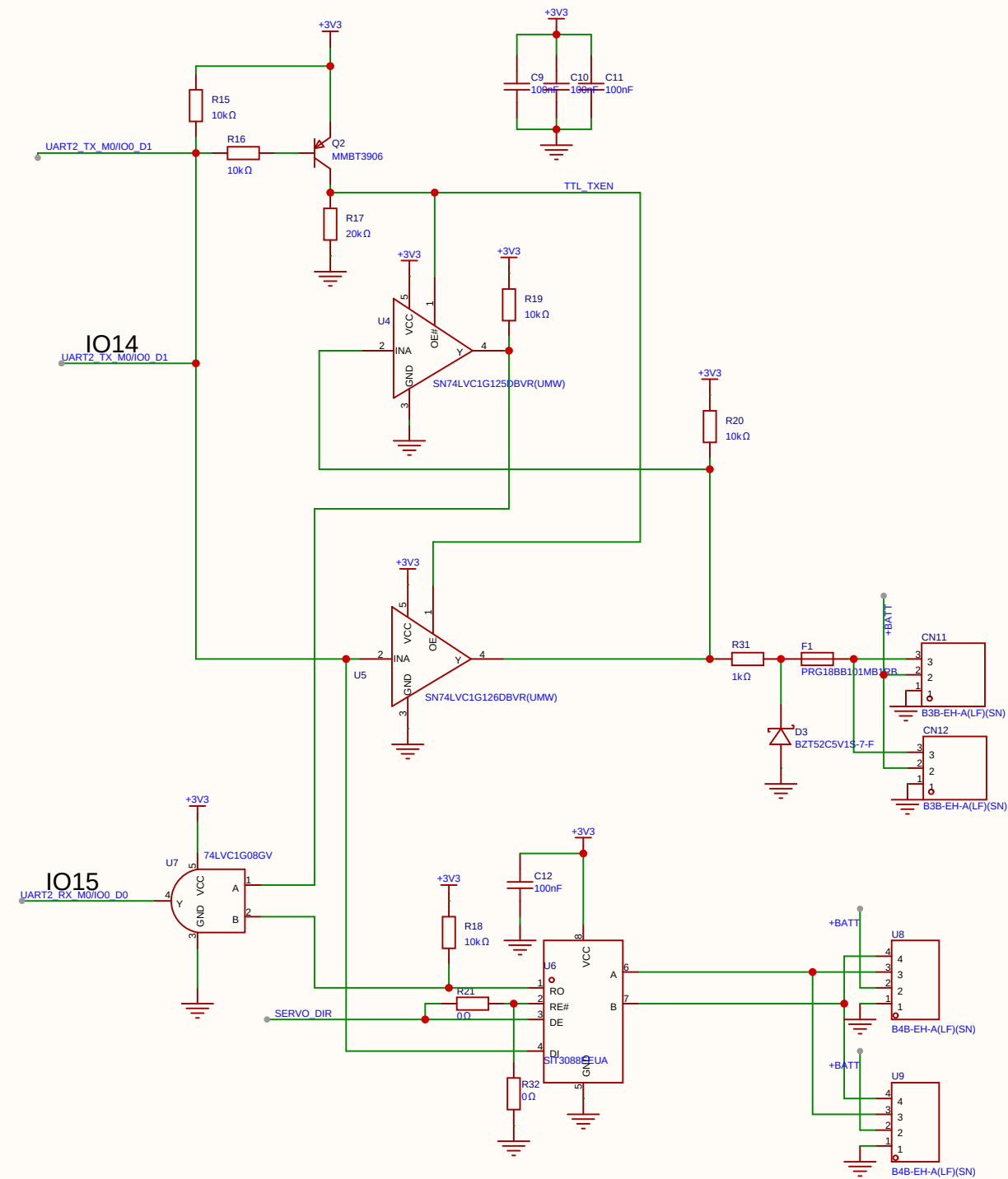


I2C add.:
- acc.: 0x19
- gyr.: 0x68



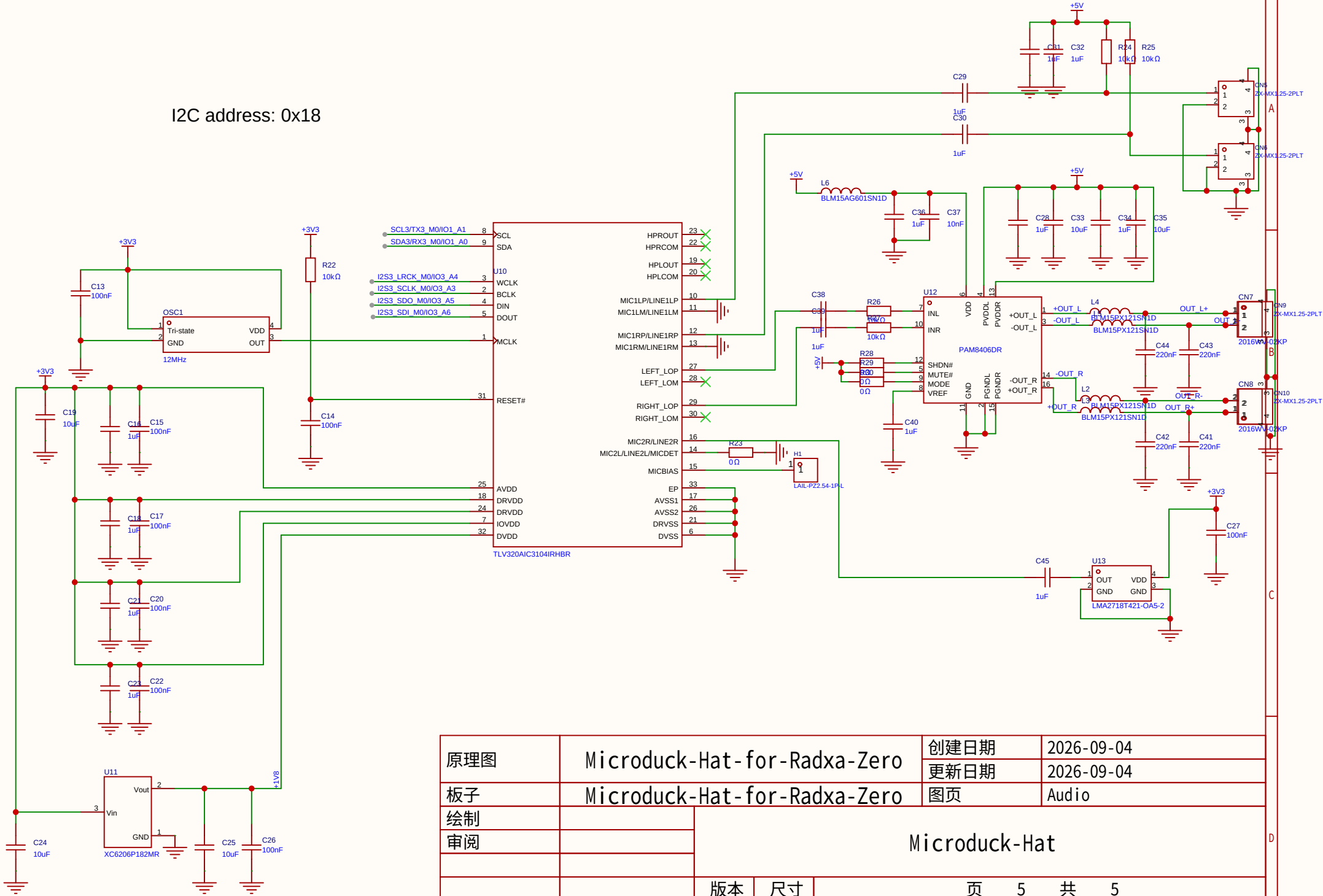
原理图	Microduck-Hat-for-Radxa-Zero			创建日期	2026-09-03
				更新日期	2026-09-12
板子	Microduck-Hat-for-Radxa-Zero			图页	Sensor
绘制		Microduck-Hat			
审阅					
		版本	尺寸	页 3 共 5	
		V1.0	A4	嘉立创EDA	

SERVOS



原理图	Microduck-Hat-for-Radxa-Zero			创建日期	2026-09-03
				更新日期	2026-09-20
板子	Microduck-Hat-for-Radxa-Zero			图页	Servo
绘制		Microduck-Hat			
审阅					
		版本	尺寸	页 4 共 5	
		V1.0	A4	嘉立创EDA	

I2C address: 0x18



原理图	Microduck-Hat-for-Radxa-Zero			创建日期	2026-09-04		
				更新日期	2026-09-04		
板子	Microduck-Hat-for-Radxa-Zero			图页	Audio		
绘制		Microduck-Hat					
审阅							
		版本	尺寸	页	5	共	5
		V1.0	A4	嘉立创EDA			